

Scratchwork and Studies related to TFIM

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ABSTRACT: First of all we look at the *Transverse Field Ising Model*.

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1 The Transverse Field Ising Model

For now we will work with Spin $\frac{1}{2}$ particles. The spin of the particles are represented by a ket $|\uparrow\rangle = |1\rangle \in \mathcal{H}, |\downarrow\rangle = |0\rangle \in \mathcal{H}$ where $\mathcal{H}$ is a Hilbert Space for spin $\frac{1}{2}$.

Let's construct a 1+1d lattice with each site labeled as j ranging from $j = 1$ to $j = L$. Each site j of the lattice is assigned to a Hilbert Space $\mathcal{H}_j$. The Hilbert Space for the lattice $\mathcal{H}$ is defined by

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \cdots \otimes \mathcal{H}_L = \bigotimes_{j=1}^L \mathcal{H}_j \quad (1.1)$$

Definition 1: Transverse Field Ising Model

The Hamiltonian for the lattice $H : \mathcal{H} \rightarrow \mathcal{H}$ is defined by

$$H = - \sum_{i=1}^L (Z_{i-1} Z_i + X_i)$$

where Z_i, X_i are Pauli Matrices, in $|1\rangle, |0\rangle$ basis given by:

$$Z_i = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}_{i\text{-th site}} \quad \text{and} \quad X_i = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}_{i\text{-th site}}$$

2 Dummy Introduction

Theorem 1: Fluid Equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad \text{and} \quad \rho \frac{D\mathbf{u}}{Dt} = -\nabla P$$

and using Theorem [1](#)

References

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